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Working Group VII - Dark Matter

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Summary

This working group will focus on the effects of baryonic physics on the properties of the simulated galaxies' dark matter halos.

Participants

Project Leaders: Miguel Rocha

Kenza Arraki, Mike Boylan-Kolchin, Javiera Guedes, Sam Leitner, Piero Madau, Annalisa Pillepich, Joel Primack, Justin Read, Miguel Rocha, Romain Teyssier, Sebastian Trujillo-Gomez, Adi Zolotov

Please email Miguel Rocha (rocham@uci.edu) if you would like to participate in this working group.

Scientific Goals

Questions we will address include (but are not limited to) the following:

1. Is there evidence for the formation of DM cores, and how much does it depend on numerical resolution, feedback implementation and galaxy properties?
2. How much adiabatic contraction is there, and is it well described by existing analytical prescriptions? How does this depend on time, and on the mass and accretion history of the halo?
3. Is there evidence for a dark disk, and how does it depend on merger history?
4. Is the substructure abundance depleted or enhanced, and how is its distribution in the host halo altered?
5. Does baryonic mass accretion track DM mass accretion? If not, how and why does it differ?
6. How much does baryonic physics alter the shapes of halos? As a function of radius?
7. Do we see evidence for a DM density displacement from the galaxies' dynamical center?

Where possible we will try to directly connect our findings to implications for astronomical observations and DM detection experiments.

Simulations

We currently have representatives from the following simulation codes in our group: Enzo, Gadget, Gasoline, ART-I, ART-II, and RAMSES.

In order to study core formation and baryon contraction in the inner regions of DM halos we will need to resolve the

density profiles as far in as $r \sim 0.05 \cdot r_s$ (r_s = NFW scale radius). Using NFW profiles and the density version of Eq. 20 in Power et al. (2003), I found that $\sim 1e7$ particles within the virial radius are necessary to resolve these scales (this result is very insensitive to M_{vir}), cored halos are even harder to resolve since the 2-body relaxation time goes as the $\sqrt{\text{avg. enclosed density}}$ for a fixed volume and particle mass.

Based on the above, simulations from the following ICs will be needed:

$M_{vir} \sim 1e10$ (quiescent & active) with $l_{max} = 10$ ($mp \sim 1e4$) and 11 ($mp \sim 1e3$)

$M_{vir} \sim 1e11$ (quiescent & active) with $l_{max} = 12$ ($mp \sim 3.4e5$) and 13 ($mp \sim 4e4$)

$M_{vir} \sim 1e12$ (quiescent & active) with $l_{max} = 12$ ($mp \sim 3.4e5$), 13 ($mp \sim 4e4$) and 14 ($mp \sim 5e3$)*

*Level 14 would be necessary if we want to study subhalos with $M_{vir} \sim 1e10$ in a MW like host

List of Planned Simulations

Gadget3 (Resp: Miguel Rocha)

- Dark matter only simulations
- Gadget3 + Hopkins feedback + Hopkins SPH

Gasoline (Resp: Annalisa Pillepich)

- Dark matter only simulations
- Gasoline + Blastwave

Art-I (Resp: Kenza or Sebastian?)

- Dark matter only simulations
- Art-I + rad. pressure feedback

Progress

DM-only simulations with P. Hopkins' Gadget3:

Resolution Halo	$mp \sim 1.3e4$ (L10)	$mp \sim 1.6e3$ (L11)
10q	Done	Done
10v	Done	Done

Resolution Halo	$mp \sim 3.4e5$ (L12)	$mp \sim 4.2e4$ (L13)
11q	Done	Done
11v		

Resolution Halo	$mp \sim 3.4e5$ (L12)	$mp \sim 4.2e4$ (L13)
12q	Done	Running
12v		

I will add links to visualizations, profiles and subhalo mass functions plots for the completed runs as they become available.

Working Group VII - Blog

Progress? Any progress on the DM-only simulations for codes other than Gadget?

Posted Dec 13, 2013, 5:11 AM by Miguel Rocha

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